Civics Group	Reg Number

Candidate Name ________________________________

MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations
Higher 2

H2 Biology 9648/1

Paper 1

22 **September 2011**

1 hour 15 min

Additional Materials: OMR Sheet

Instructions to Candidates:

Do not open this booklet until you are told to do so.

Write your name, civics group and CG register number on the OMR and on this booklet.

Shade you CG registration number on the OMS.

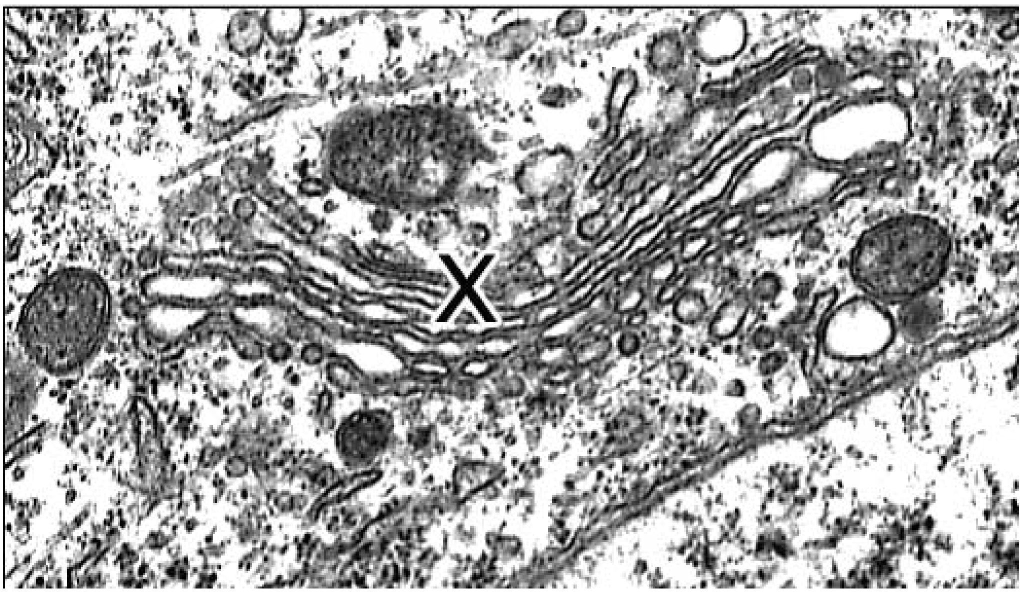
There are **forty** questions in this part. Answer **all** questions. For each question, there are four possible answers labelled **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate **OMR** answer sheet.

This document consists of **24** printed pages.

1. Which combination would decrease the fluidity of a cell surface membrane?

	saturated fatty acids	non-saturated fatty acids	cholesterol
A	decrease	increase	decrease
B	decrease	increase	increase
C	increase	decrease	decrease
D	increase	decrease	increase

2. The electron micrograph below depicts organelle X.



Which of the following options is not a function of organelle X?

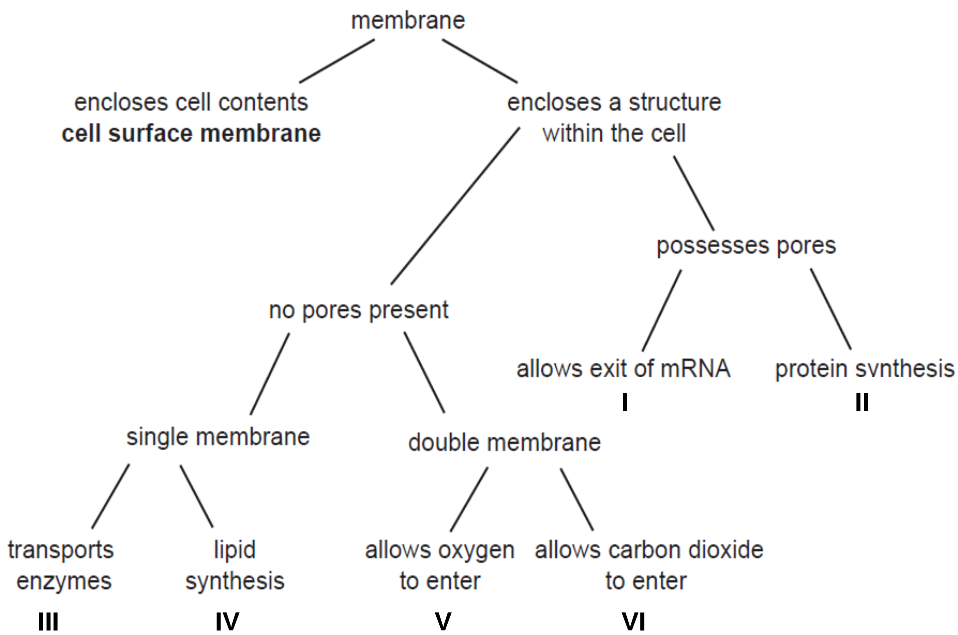
A. Glycosylation

B. Regulation of cellular calcium ion concentrations

C. Formation of secretory vesicles

D. Formation of lysosomes

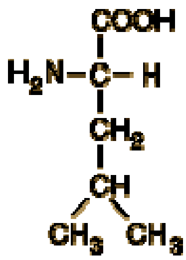
3. Membranes within and at the surface of cells have different roles. The flowchart shows a scheme to identify various organelles within the cell, by describing the structure and function of their membranes.



Which of the following outcomes correctly identifies the various cellular organelles?

	I	II	III	IV	V	VI
A	chloroplast	vesicle	smooth ER	rough ER	nucleolus	mitochondrion
B	nucleolus	rough ER	vesicle	smooth ER	nucleus	mitochondrion
C	nucleus	smooth ER	mitochondrion	rough ER	vesicle	chloroplast
D	nucleus	rough ER	vesicle	smooth ER	mitochondrion	chloroplast

4. The diagram shows a molecule which is an important component of living organisms.

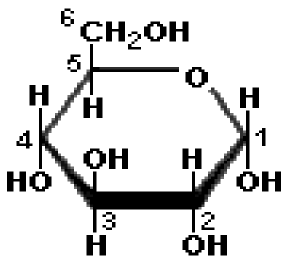


A forensic sample is tested for the presence of a polymer of this molecule.

Which polymer is tested for?

A. DNA
B. Triglyceride
C. Protein
D. Starch

5. A glucose molecule in the diagram has its six carbon atoms numbered.

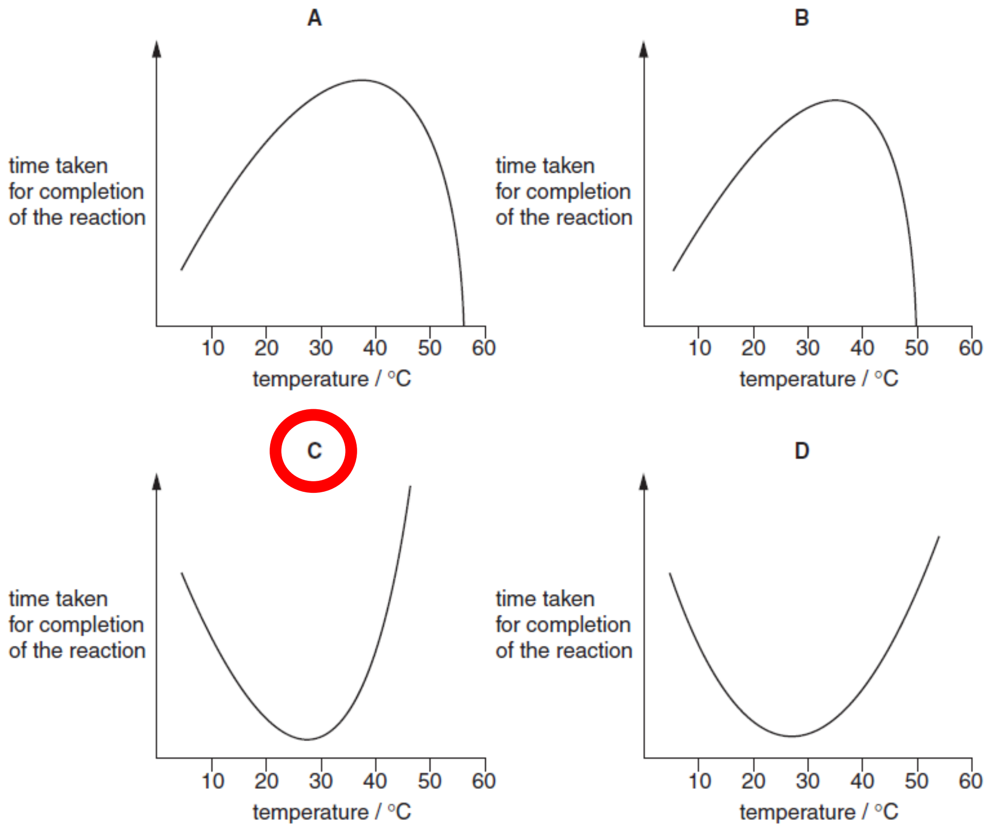


Which of the following statement is false?

A. Carbon 1 and 4 are involved in the formation of amylase
B. Carbon 1 and 6 are involved in the formation of amylopectin
C. Carbon 1 and 4 are involved in the formation of cellulose
D. Carbon 1 and 6 are involved in the formation of starch

6. An enzyme is completely denatured at 50°C. A fixed concentration of this enzyme is added to a fixed concentration of its substrate. The time taken for completion of the reaction is measured at different temperatures.

Which graph shows the results?



7. Which correctly describes the structure of DNA during transcription?

A. Double-stranded with single-stranded sections associated with polymerase enzymes.

B. Double-stranded region with two loops of unpaired bases.

C. Double strand held together by covalent bonds between nucleotide pairs.

D. Single strand formed by base pairing cytosine with guanine and adenine with thymine.

8. A student obtained a sample of DNA. mRNA was transcribed from this DNA. He then separated the two strands of the DNA sample by adding NaOH. The base compositions of each strand and that of the mRNA were analysed. The results of the analysis are shown in the table below.

	A	**G**	**C**	**T**	**U**
DNA strand 1	19.1	26.0	31.0	23.9	0.0
DNA strand 2	24.2	30.8	25.7	19.3	0.0
DNA strand 3	20.5	25.2	29.8	24.5	0.0
mRNA	19.0	25.9	30.8	0.0	24.3

Which strand of DNA serves as a template for mRNA synthesis?

A. Strand 1
B. Strand 2
C. Strand 3
D. Strands 2 and 3

9. Which of the following observations proves that the anticodon of a tRNA molecule, and not the amino acid that it carries, recognizes and binds to the mRNA codon at the ribosome?

A. A tRNA carrying a valine but with an isoleucine anticodon does not place any amino acid onto a growing peptide chain when an isoleucine mRNA codon is present.
B. A tRNA carrying a valine with a valine anticodon places a valine onto a growing peptide chain when only the first two bases of the anticodon pairs with the mRNA codon.
C. A tRNA carrying a valine with a valine anticodon does not place any amino acid onto a growing peptide chain when only the first two bases of the anticodon pairs with the mRNA codon.
D. A tRNA carrying a valine but with an isoleucine anticodon places a valine onto a growing peptide chain when an isoleucine mRNA codon is present.

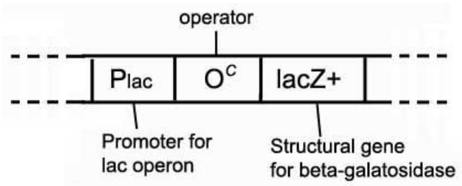
10. A bacterium can make or absorb the amino acid glycine from its surroundings. It is found that glycine binds and activates a repressor protein of an operon. The presence of glycine will result in

A. the breakdown of glycine.
B. inhibition of the synthesis of glycine.
C. inhibition of repressor protein synthesis.
D. manufacture of the repressor protein.

11. Which process in bacteria always allows chromosomal and non-chromosomal DNA to be transferred?

A. Binary fission
B. Conjugation
C. Transduction
D. Transformation

12. *The lac operon of an E. coli* strain carries the P_{lac} O^c *lacZ*+ DNA sequence as shown below. In this sequence, the *lacZ* allele is normal (lacZ+) but the operator (O^c) is mutated. The O^c mutant is unable to bind the *lac* repressor.



How would the expression of the *lacZ* gene be affected in this strain of cells?

A. There will be normal regulation of lactose metabolism.
B. There will be continuous expression of beta-galactosidase regardless of glucose levels.
C. The inability to synthesis the *lacZ* gene product, beta-galactosidase.
D. The *lacZ* gene is inducible, but unable to be repressed by high glucose levels.

13. When DNA is compacted by histones into 10 nm and 30 nm fibers, the DNA is unable to interact with proteins required for gene expression. Therefore, to allow for these proteins to act, the chromatin must constantly alter its structure. Which processes contribute to this dynamic activity?

A. DNA supercoiling at or around histones
B. acetylation of histone proteins
C. hydrolysis of DNA molecules where they are wrapped around the nucleosome core
D. uncoiling of euchromatin by transcription factors

14. The hepatitis C virus is a single-stranded, enveloped, positive sense RNA virus, which attack the liver cells of human beings. Which of the following general strategy states how the viruses enter the host cell?

A. uncoating of the genetic material of the virus via endocytosis
B. fusion of the viral envelope with the host cell membrane
C. injection of the whole virus into the host cell
D. penetration of the viral capsid via budding

15. The H1N1 virus is a subtype of the influenza A virus. Which statement about the H1N1 virus is true?

A. The virus contains the enzyme reverse transcriptase.
B. The virus comprises of a nucleocapsid enclosed by a membrane.
C. The virus is not encased by a lipoprotein envelope.
D. The virus genome is single-stranded DNA.

16. Four different genes are regulated in different ways.

- Gene 1 undergoes tissue-specific patterns of alternative splicing.
- Gene 2 is part of a group of structural genes controlled by the same regulatory sequence.
- Gene 3 is in some circumstances subject to methylation
- Gene 4 codes for a repressor protein which acts at an operator site close by.

Which row of the table correctly identifies which genes are prokaryotic and which are eukaryotic?

	Prokaryotic	**Eukaryotic**
A.	1 and 2	3 and 4
B.	1 and 3	2 and 4
C.	2 and 3	1 and 4
D.	2 and 4	1 and 3

17. Messenger RNA is isolated by passing homogenised tissue through a fractionating column. The column consists of short lengths of uracil nucleotides attached to a solid support medium. The mRNA that passes through the column cannot be translated as they break up into fragments. The mRNAs that attach to the column can be separated by application of sodium hydroxide.

Which statement is true of the mRNA in this experiment?

(i) Complementary base-pairing occurs between adenine and uracil bases in the column.

(ii) The mRNA that pass through the fractionating column probably lack poly A tails.

(iii) The mRNA that attach to the fractionating column can be translated.

(iv) The mRNA that attach to the fractionating column probably lack poly A tails.

A. (i) and (iv)

B. (i), (ii) and (iii)

C. (ii) and (iv)

D. All of the above.

18. Which of the following groups is/are not example(s) of oncogene products?

(i) Second messengers

(ii) Cell surface receptors

(iii) GTPase involved with cellular signaling

(iv) Cyclins that prepare the cell to undergo mitosis

A. (i) only

B. (i) and (iii)

C. (i), (ii) and (iv)

D. All of the above.

19. In rats, the allele of a gene for 'mottled' coat (M) and the recessive allele (m) for 'normal' coat are sex-linked. The allele of a gene for 'long' whiskers (W) and the recessive allele (w) for 'short' whiskers are autosomal.

A male rat with a normal coat and short whiskers was mated on several occasions to the same female. The offspring showed the following phenotypes in equal proportions.

- Mottled females and males with long whiskers
- Mottled females and males with short whiskers
- Normal females and males with long whiskers
- Normal females and males with short whiskers

What are the genotypes of the parents?

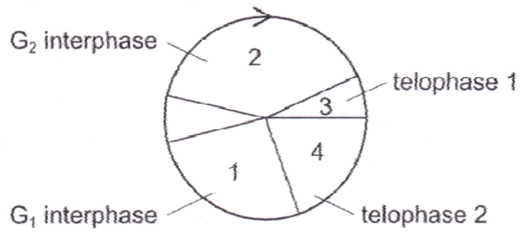
A. X^mYww and X^MX^MWW

B. X^mY^mww and X^MX^MWw

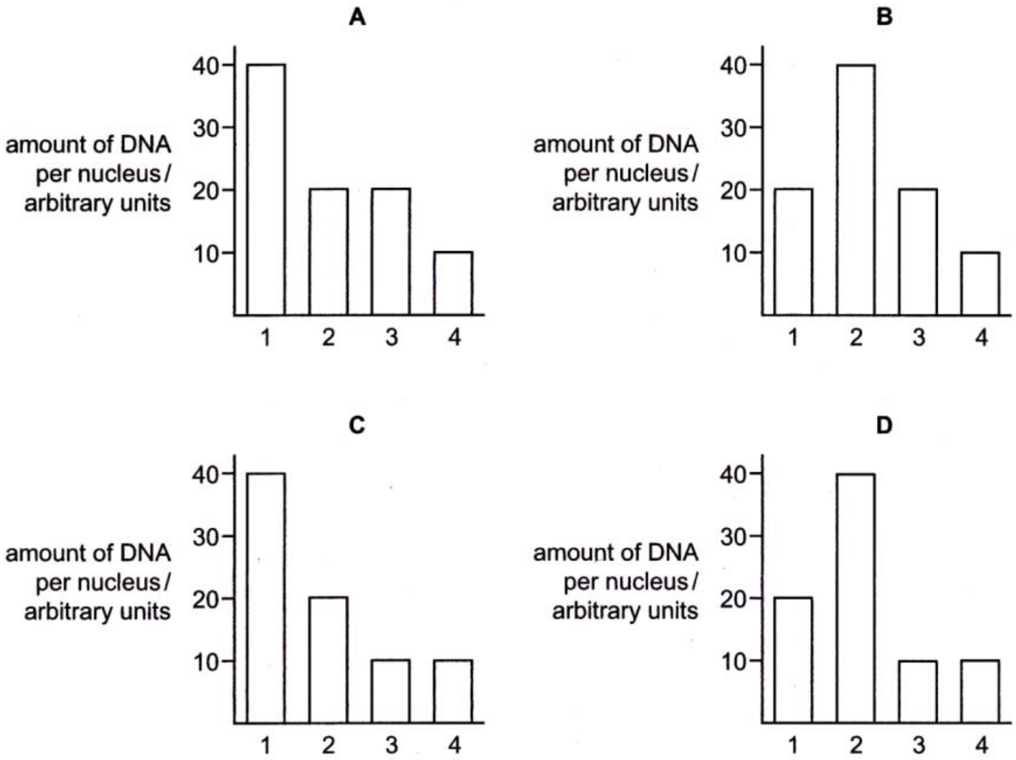
C. X^mY^mww and X^MX^mWW

D. X^mYww and X^MX^mWw

20. The amount of DNA per nucleus in a cell during a meiotic cell cycle, as shown below was measured.

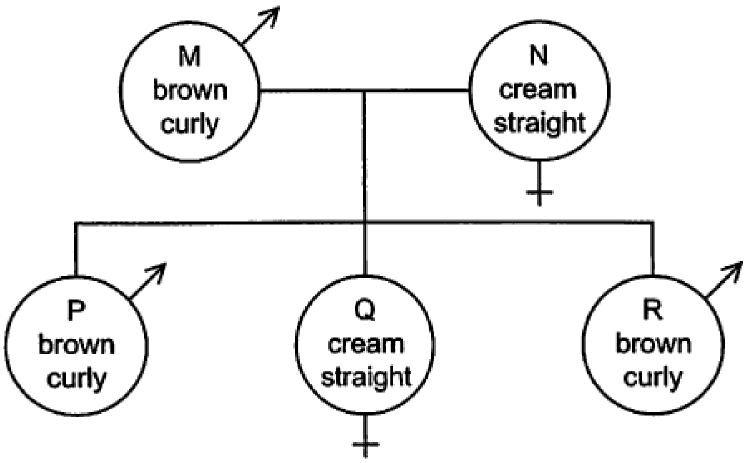


Which bar graph correctly represents the variation in the DNA content?



21. Assume that in goats, a pair of alleles is responsible for the inheritance of hair color and that another pair controls hair texture. These pairs are located on different autosomes. The allele for brown hair (B) is dominant to the allele for cream hair (b), and the allele for curly hair (C) is dominant to the allele for straight hair (c).

    The diagram shows a cross between two goats.



If R is mated with a female goat of the same genotype as M, what are the chances of the first offspring being a male with cream colored, straight hair?

A. 0
B. 1 in 32
C. 1 in 16
D. 1 in 4

22. In *Drosophilia melanogaster*, the gene loci for the recessive allele for pink eyes and the recessive allele for blistered wings are next to each other on the same chromosome.

A pure-breeding fly with red eyes and long wings is crossed with a fly with pink eyes and blistered wings.

The F1 generation all had red eyes and long wings. Two of the F1 were crossed and produced 416 offspring.

Which shows the numbers of each phenotype in the F2 generation?

	Phenotype			
	red eyes and long wings	red eyes and blistered wings	pink eyes and long wings	pink eyes and blistered wings
A.	226	46	44	100
B.	312	0	0	104
C.	338	0	0	78
D.	232	78	82	24

23. Two genes involved in coat color of goats are at loci on different chromosomes.

The color gene C causes the hairs to be uniform color and has three alleles.

- C^{DB} giving dark brown hairs
- C^{B} giving black hairs
- C^{MB} giving medium brown hairs

A dominant allele of the agouti gene (A^{G}) causes the development of white hairs between the colored hairs giving the coat a shaded appearance.

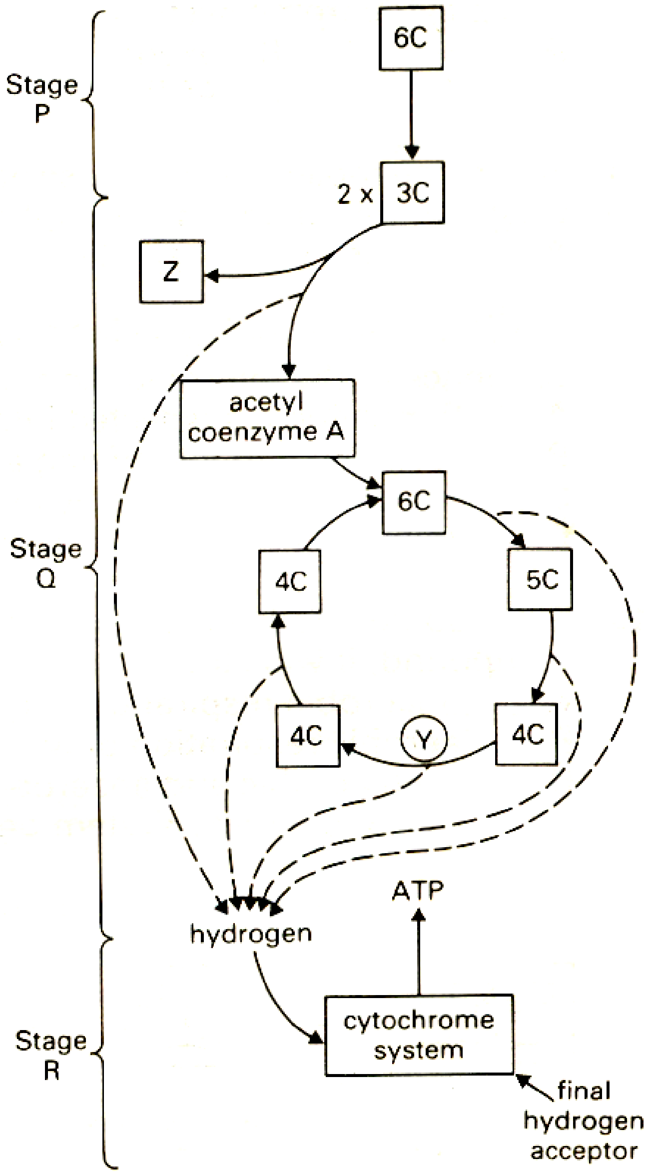
The table shows the results of crosses between a male goat and two female goats.

Cross	Parents	Offspring
1	Black agouti male **x** Uniformly dark brown female	50% uniform, 50% agouti; 50% dark brown, 50% black
2	Black agouti male **x** Medium brown agouti female	all agouti; 50% black, 50% medium brown

Which row shows the possible genotypes of the male and female goats?

	Black agouti male	Uniformly dark brown female	Medium brown agouti female
A.	$C^{B}C^{MB}A^{G}A^{G}$	$C^{B}C^{DB}A^{G}A^{g}$	$C^{MB}C^{MB}A^{G}A^{G}$
B.	$C^{B}C^{MB}A^{G}A^{g}$	$C^{B}C^{DB}A^{g}A^{g}$	$C^{B}C^{MB}A^{G}A^{G}$
C.	$C^{B}C^{DB}A^{G}A^{G}$	$C^{DB}C^{MB}A^{G}A^{g}$	$C^{B}C^{DB}A^{G}A^{g}$
D.	$C^{B}C^{MB}A^{G}A^{g}$	$C^{B}C^{DB}A^{g}A^{g}$	$C^{MB}C^{MB}A^{G}A^{G}$

24. The diagram shows a biochemical pathway in a typical eukaryotic cell.



Which of the following indicates the correct locations at which stages P, Q and R occur?

	Cristae	Cytosol	Mitochondrial matrix
A.	R	P	Q
B.	Q	P	R
C.	P	R	Q
D.	R	Q	P

25. The Calvin cycle can be divided into three phases, involving several reactions.

1. carbon dioxide uptake
2. carbon reduction
3. regeneration

Which of the following correctly matched the reactions with three phases?

	ATP hydrolysis	Formation of NADP	Formation of 3C sugar	Use of RuBP
A.	1	3	2	1
B.	1, 3	2	1, 2	3
C.	2	1, 2	2, 3	3
D.	2, 3	2	2	1

26. Which of the following processes release ATP in a plant cell during daytime?

I oxidative phosphorylation
II photophosphorylation
III substrate level phosphorylation
IV oxidative decarboxylation
V hexose phosphorylation

A. I, III and V

B. II, III and IV

C. I, II and V

D. I, II and III

27. Which of the following statements are true about the signal transduction pathways?

I They enable different cells to respond appropriately to the same signal
II They help cells use up phosphate generated by ATP breakdown.
III They can amplify a signal
IV A single ligand can lead to different signal transduction pathways, resulting in different cellular responses.

A. I, II and III

B. I, III and IV

C. II and IV

D. I and III

28. The hormone insulin binds to the tyrosine kinase receptors. Which of the following shows the correct sequence of events following the binding of insulin to the receptor?

I. phosphorylation of tyrosine kinases
II. dimerization of tyrosine kinase receptors
III. full activation of tyrosine kinases.
IV. activation of relay proteins
V. binding of relay proteins to phosphotyrosines

A. II $\rightarrow$ III $\rightarrow$ I $\rightarrow$ V $\rightarrow$ IV

B. IV $\rightarrow$ V $\rightarrow$ II $\rightarrow$ I $\rightarrow$ III

C. II $\rightarrow$ I $\rightarrow$ III $\rightarrow$ V $\rightarrow$ IV

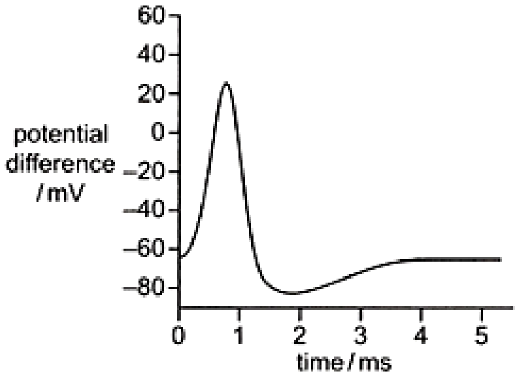
D. V $\rightarrow$ IV $\rightarrow$ II $\rightarrow$ I $\rightarrow$ III

29. Insulin is released from cells in the islets of Langerhans while acetylcholine is released from neurons. Both events take place in response to changes in cytosolic concentration of calcium ions.

Which of the following correctly describes the release of insulin and acetylcholine?

	Release of insulin	Release of acetylcholine
A.	Arrival of an action potential causes Ca^{2+} to enter the cytosol of β-cells	Signal transduction causes Ca^{2+} to enter the cytosol of neurons
B.	Signal transduction causes Ca^{2+} to enter the cytosol of β-cells	Arrival of an action potential causes Ca^{2+} to enter the cytosol of neurons
C.	Arrival of an action potential causes Ca^{2+} to exit the cytosol of α-cells	Signal transduction causes Ca^{2+} to exit the cytosol of neurons
D.	Signal transduction causes Ca^{2+} to exit the cytosol of α-cells	Arrival of an action potential causes Ca^{2+} to exit the cytosol of neurons

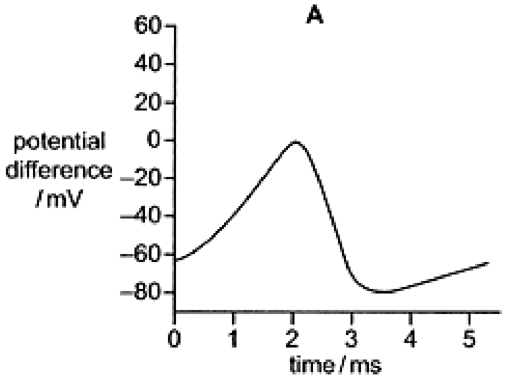
30. The diagram shows a normal action potential.



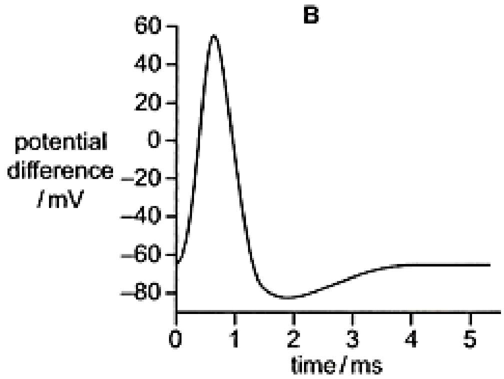
Tetraethylammonium ions bind with and block voltage-gated K^+ channels.

Which diagram shows an action potential in a neuron affected by a small quantity of tetraethylammonium ions?

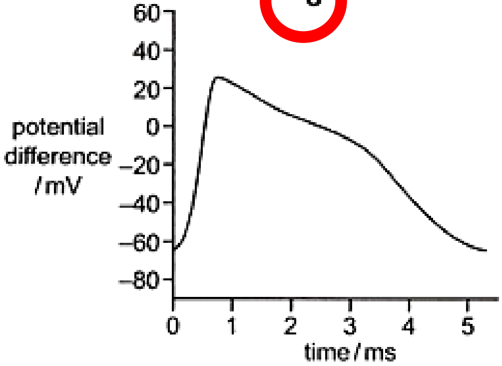
A



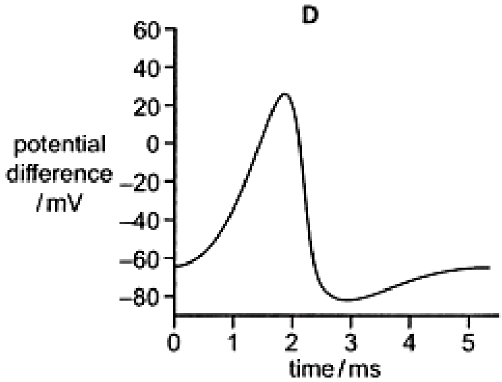
B



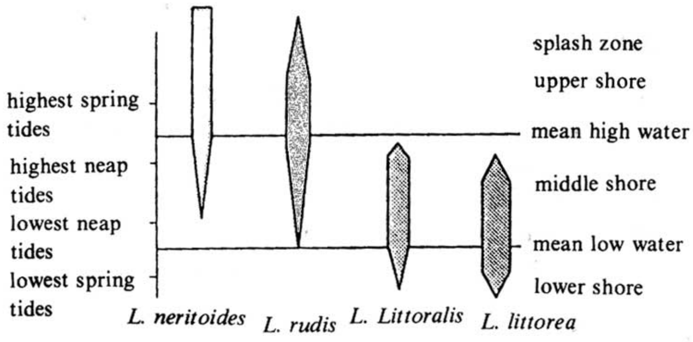
C



D



31. The diagram below shows the frequency and distribution of four *Littorial* species (periwinkles) on a rocky shore. All feed in a snail-like manner by grazing on algae.



Which of the following factors could not directly contribute to this distribution pattern?

A. Variation in the tolerance of each species to desiccation.

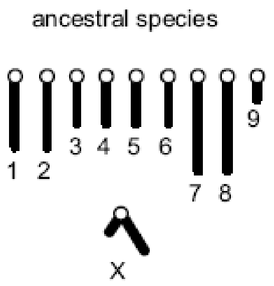
B. Variation in the tolerance of each species to wave action.

C. Photoperiod and seasonal change in day length.

D. Differential selection of periwinkles by predators.

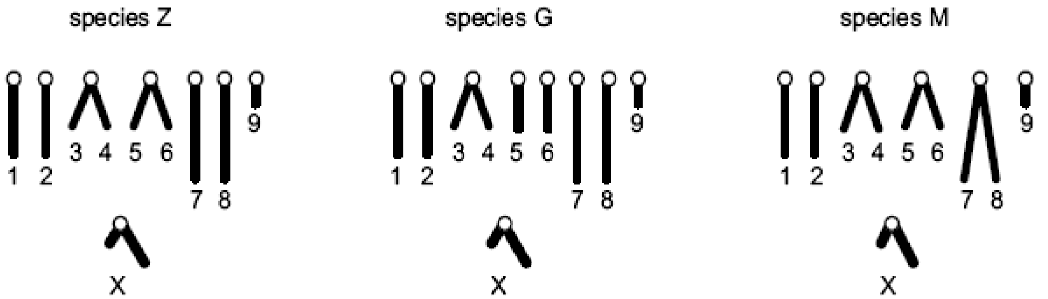
32. In some Australian insects, new species have arisen through changes that occurred to chromosomes in an ancestral species. Such changes may involve the joining together of chromosomes, the loss of whole or parts of chromosomes, and rearrangement of the genetic material within chromosomes.

One ancestral species has the following haploid set of chromosomes.



As the changes in chromosomes accumulate, a number of different species can result from a single ancestral species.

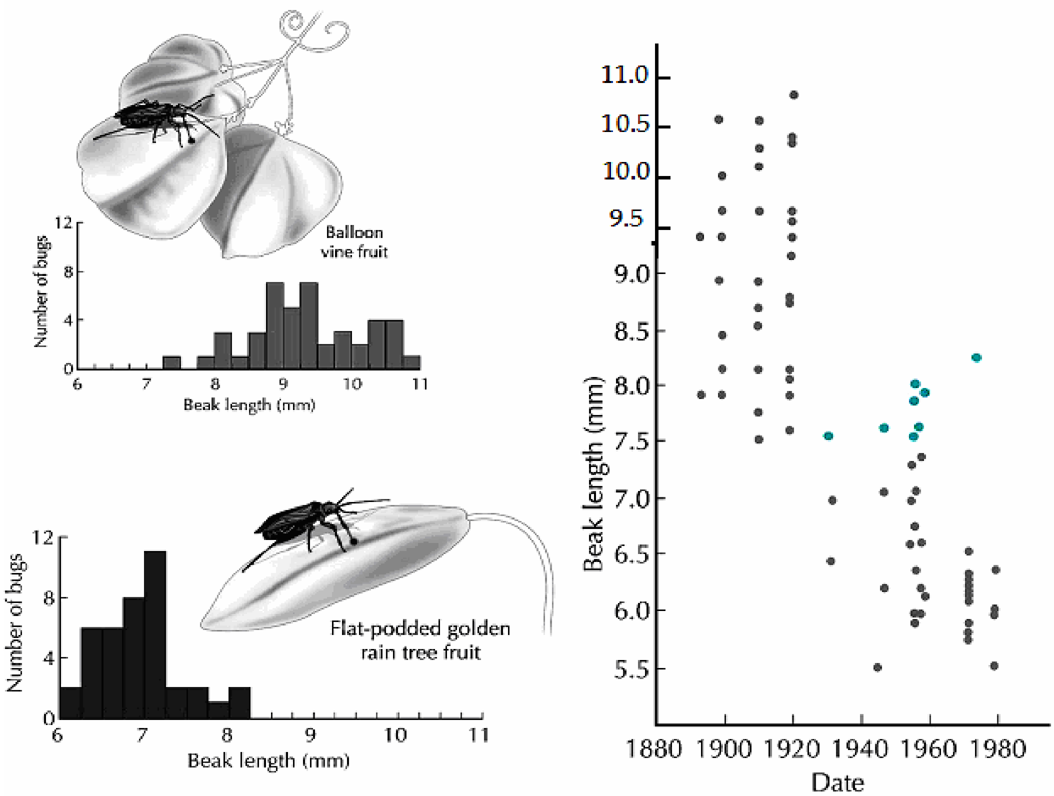
Three species that have evolved from the ancestral species shown above have the haploid sets of chromosomes shown below.



The most likely order of evolution of these species is

A. ancestral species, species Z, species G, species M.
B. ancestral species, species Z, species M, species G.
C. ancestral species, species G, species Z, species M.
D. ancestral species, species G, species M, species Z.

33. Soapberry bugs in America originally fed on the native balloon vine using their sharp beaks to penetrate the fruit. In the 1920s, the flat-podded golden rain tree was introduced from Asia. This tree produces thinner-skinned fruit.



What can you conclude from the data presented above?

I Soapberry bugs with shorter beaks are being selected for.

II Soapberry bugs developed shorter beaks after they switched to feed on this new host plant.

III Soapberry bugs had a preference for the introduced flat-podded golden rain tree fruits.

IV Soapberry bugs population contained the allele for short beaks since 1900.

A. I only

B. I and II only

C. I, II and III

D. II, III and IV

34. In humans, the CFTR gene codes for a transmembrane protein needed for the diffusion of Cl^- ions into and out of epithelial cells. More than 1000 types of mutation of the CFTR gene has been discovered.

Which of following statements are correct?

1. CFTR gene is polymorphic
2. The gene locus for CFTR is multiple allelic.
3. The gene locus for CFTR is polygenic
4. Allele frequency of the CFTR gene in human population is more than 1000

A. 1 and 2 B. 2 and 3 C. 1 and 4 D. 3 and 4

35. Which of the descriptions are correct?

	cDNA library	Genomic library
A.	A collection of clones of overlapping DNA fragments representing the entire genome	A collection of genes that may be used in genetic engineering
B.	A collection of DNA fragments from known chromosomal locations	A collection of genes without introns
C.	A collection of the DNA of expressed genes	A collection of clones whose relationship to each other can be established by genomic mapping
D.	A collection of the coding and non-coding DNA of a single species	A collection of genes of a single chromosome

36. Single stranded primers are added to the reaction mixture of polymerase chain reaction (PCR) and the mixture heated to 95°C followed by cooling to 55°C.

Which statement explains why DNA denatured at 95^{o}C anneal with primers but not to each other?

A. The primers are shorter and thus collide with the denatured DNA at higher frequency.

B. The primers anneal to 3’ end of the denatured DNA.

C. The primer concentration exceeds the denatured DNA concentration.

D. The denatured DNA is mutated by heat and is no longer complementary to each other.

37. The restriction enzyme, *Sau* 3A, cuts DNA as shown.

- - - - G A T C - - - -		- - - -	G A T C - - - -
- - - - C T A G - - - -	→	- - - - C T A G	- - - -

Which of the plasmids below, with specific regions indicated, can be cut using restriction enzyme *Sau* 3A in order to allow DNA to be inserted into the circular plasmid without loss of any part of the plasmid?

A.	- - - - T A G A T C A C - - - - C T C A A G C T - - - - - - - - A T C T A G T G - - - - G A G T T C G A - - - -
B.	- - - - T T A G A T C A - - - - C G A T C G G T - - - - - - - - A T T C T A G T - - - - G C T A G C C A - - - -
C.	- - - - G A T C G A T C - - - - T C C A C G T A - - - - - - - - C T A G C T A G - - - - A G G T G C A T - - - -
D.	- - - - A A T C A G G T - - - - C C A C C T G A - - - - - - - - T T A G T C C A - - - - G G T G G A C T - - - -

38. Which of the following processes could not improve quality of the crop?

W inserting genes for insect pest resistance into potato

X inserting genes for vitamin A production into rice

Y inserting genes for nitrogen fixation into soya beans

Z inserting genes for drought resistance into rice

A. W and X B. X and Y C. Y and Z D. Z and W

39. Scientists are concerned about the escape of genetically modified salmon into the wild.

What is the most likely reason for this concern?

A. The genetically modified salmon may not survive in the wild.
B. The mutation rate of the genetically modified salmon will increase.
C. The genetically modified salmon may replace the wild salmon population.
D. The growth rate of the genetically modified salmon will be affected.

40. Some of the goals of the Human Genome Project are:
- To determine the sequence of the entire human genome
- To identify all the genes in the human genome
- To find the locus of all the genes on the 46 human chromosomes

Which of the following are ethical concerns arising from the goals stated?

1. Anthropologist tracing the ancestry of human populations.
2. Parents choosing embryos for implantation only after tests for acceptable genes.
3. Insurance company offering cheaper rates to people with genetic disposition to fewer diseases.
4. Scientists developing tests for only some disease-causing genes.
5. Genetic counselors giving advice to people who are genetically pre-disposed to risks.

A. 2 and 3 B. 3 and 4 C. 1 and 5 D. 4 and 5

End of Paper.